

Dear Hiring Team,

I am writing to express my strong enthusiasm for the DSP Firmware Engineering Co-op/Intern position for the Winter 2027 term in Ottawa. As a Software Engineering Co-op student at McGill University with prior engineering experience in the public sector and research, I am excited about the opportunity to contribute to Infinera's next-generation Optical Modem ASIC and embedded control systems.

Through my academic curriculum and software engineering projects, I have developed solid proficiency in modern C++, object-oriented design, and algorithm optimization. Complementing my software background, my research work at the National Institute for Scientific Research (INRS) and technical role at the Ministry of Transportation involved writing performance-critical code in C++ and Python, performing system simulations, and handling complex dataset processing. I am comfortable using modern development tools including Git, CMake, and CI/CD concepts to maintain clean, scalable, and fully documented code bases.

In addition to my C++ background, I bring strong mathematical and simulation capabilities. My experience developing custom algorithms and modeling data systems aligns well with performing MATLAB simulations, hardware modeling, and digital system validation. Furthermore, my hands-on background in hardware labs and network environments including building custom server infrastructure and configuring automated pipelines gives me a practical understanding of low-level system behaviors, real-time control, and hardware-software interaction.

Having completed previous engineering work terms at the Ministry of Transportation and INRS, I understand the rigor required in professional engineering environments. My strong problem-solving skills, adaptability, and dedication to program success will allow me to integrate quickly into your firmware team and deliver meaningful results from day one.

I would welcome the opportunity to discuss how my C++ development skills, simulation background, and engineering experience align with your team's goals for this internship.

Thank you for your time and consideration.

Sincerely,

Renji Jenkins